Mid term exam Quantum Physics (GEMF), April 2024

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1. **Beam splitter**

   (a) Write down the quantum state of the photon emitted by laser 1 after passing through the beam splitter 1, which transmits 75% of the photons and reflects the rest. Check that the state is normalized.

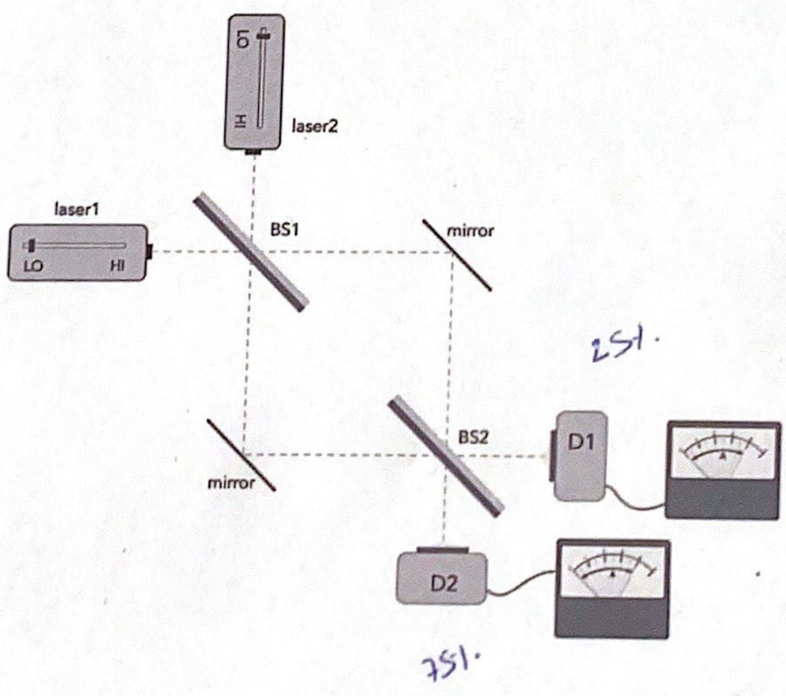
   (b) Do the same when the photon is emitted by laser 2. Check that the two quantum states are orthogonal.

   (c) Derive the matrix representation of the action of the beam splitter 1 and do the same for beam splitter 2, which transmits 25% of the photons and reflects the rest.

   (d) What is the probability that a photon emitted by laser 1 will be detected by the device D1?

   (e) What is the probability that a photon emitted by laser 2 will be detected by the device D1?

Note that the length of the two pathways between BS1 and BS2 is equal.



2. **Wave-particle duality of matter**

   (a) Calculate the wave length of a 15 mg particle that moves at a velocity 30 m/s.

   (b) Calculate the wave length of a proton moving at 1% of the speed of light in vacuum.

   (c) What is the velocity of a muon whose wave length is $1.4668 \cdot 10^{-12}$ m?

Planck's constant: $6.6261 \cdot 10^{-34}$ Js

proton mass: $1.6726 \cdot 10^{-27}$ kg

muon mass: $1.8835 \cdot 10^{-28}$ kg

speed of light in vacuum: $2.9979 \cdot 10^{8}$ m/s

3. **Formal concepts**

Let $|\phi_a\rangle = a\begin{pmatrix}2\\1\end{pmatrix}$, $|\phi_b\rangle = b\begin{pmatrix}2\\-4\end{pmatrix}$ and $\hat{A} = \begin{pmatrix}1 & i\\-i & 2\end{pmatrix}$:

(a) Show that $\hat{A}$ is an Hermitian operator.

(b) Normalize $|\phi_a\rangle$ and $|\phi_b\rangle$ and check that the two quantum states are orthogonal.

(c) Given that the quantum system is described by $|\psi\rangle = N\big(|\phi_a\rangle+|\phi_b\rangle\big)$, calculate the expectation value of $\hat{A}$.

4. **Measurements** Consider the following two operators and a quantum system that is described by the state vector ψ

$$\hat{A} = \begin{pmatrix} 1 & 3i/4 \\ -3i/4 & 3 \end{pmatrix} \qquad \hat{B} = \begin{pmatrix} 2 & \frac{1}{2} \\ \frac{1}{2} & -3 \end{pmatrix} \qquad |\psi\rangle = \frac{1}{\sqrt{5}} \begin{pmatrix} -1 \\ 2 \end{pmatrix}$$

(a) Show that the possible outcomes of measuring A are 3.25 and 0.75.

(b) Determine the probability for measuring $A = 3.25$ and $A = 0.75$.

(c) Calculate the expectation value of $\hat{B}$ after measuring $A = 0.75$.

$$\text{Eigenvectors of } \hat{A}\text{:} \quad |\phi_1\rangle = \frac{1}{\sqrt{10}} \begin{pmatrix} 1 \\ 3 \end{pmatrix} \qquad |\phi_2\rangle = \frac{1}{\sqrt{10}} \begin{pmatrix} 3 \\ -1 \end{pmatrix}$$